

Sergio Saponara

professor

基本信息

姓名: Sergio Saponara

所属院校: 比萨大学

地区: Italy

年龄范围: 50岁以上

学科领域

电子与计算机工程

人工智能

集成电路设计与集成系统

微电子科学与工程

通信工程

信息安全

电气工程及其自动化

大功率半导体科学与工程

电子信息工程

研究领域

深度学习与神经网络

嵌入式系统与硬件设计

电力电子与电机驱动

计算机视觉

密码学与密码分析

FPGA与可重构计算

低功耗电路与能量管理

片上系统与嵌入式设计

计算机体系结构

超大规模集成电路设计

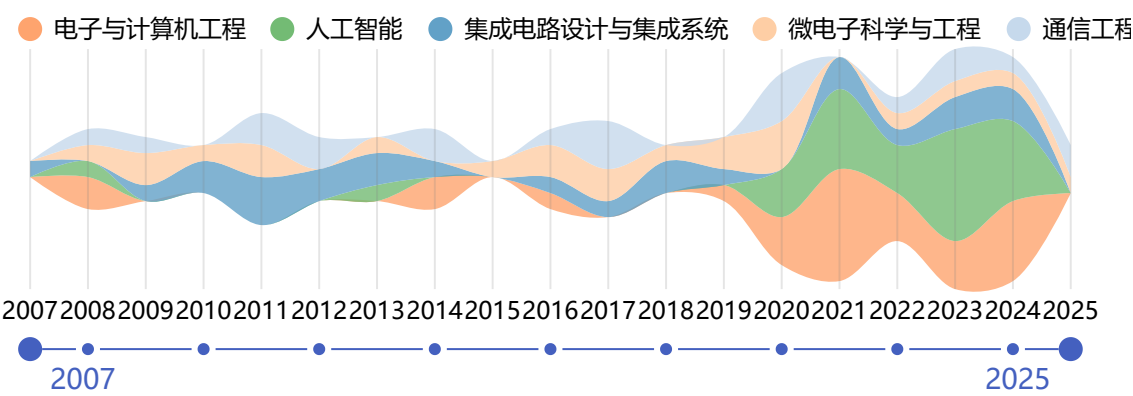
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个人简介

Prof. Sergio Saponara is a Full Professor of Electronic Engineering at the University of Pisa. He holds an IEEE Distinguished Lecturer title, a Master's and a PhD in Electronic Engineering from the same university. His research interests span integrated electronics, wireless systems, etc., with applications in vehicles, space electronics, etc. He teaches multiple courses at the University of Pisa and the Italian Naval Academy. He is involved in numerous academic and industrial activities, including founding special interest groups, serving on committees, and editing books and journals. He has received many awards and has significant technology - transfer collaborations. He is also a CTO of a spin - off

company and participates in various national and international research projects.

近年发表的论文



Cybersecurity of Automotive Wired Networking Systems: Evolution, Challenges, and Countermeasures

ELECTRONICS

Canino, Nicasio; Dini, Pierpaolo; Mazzetti, Stefano; Rossi, Daniele; Saponara, Sergio; Soldaini, Ettore

被引用次数: 0 2025

Robust automatic modulation classification using asymmetric trilinear attention net with noisy activation function

ENGINEERING APPLICATIONS OF ARTIFICIAL INTELLIGENCE

Zheng, Qinghe; Tian, Xinyu; Yu, Zhiguo; Yang, Mingqiang; Elhanashi, Abdussalam; Saponara, Sergio

被引用次数: 1 2025

Hardware Design of an Advanced-Feature Cryptographic Tile Within the European Processor Initiative

IEEE TRANSACTIONS ON COMPUTERS

Nannipieri, Pietro; Crocetti, Luca; Di Matteo, Stefano; Fanucci, Luca; Saponara, Sergio

被引用次数: 8 2025

Recent Advances in Automatic Modulation Classification Technology: Methods, Results, and Prospects

INTERNATIONAL JOURNAL OF INTELLIGENT SYSTEMS

Zheng, Qinghe; Tian, Xinyu; Yu, Lisu; Elhanashi, Abdussalam; Saponara, Sergio

被引用次数: 0 2025

A real-time constellation image classification method of wireless communication signals based on the lightweight network MobileViT

COGNITIVE NEURODYNAMICS

Zheng, Qinghe; Saponara, Sergio; Tian, Xinyu; Yu, Zhiguo; Elhanashi, Abdussalam; Yu, Rui

被引用次数: 2

2024

Testing silicon photonic Mach-Zehnder modulators versus total ionizing dose

NUCLEAR INSTRUMENTS & METHODS IN PHYSICS RESEARCH SECTION A-ACCELERATORS
SPECTROMETERS DETECTORS AND ASSOCIATED EQUIPMENT

Cammarata, Simone; Mattiazzo, Serena; Pantano, Devis; Palla, Fabrizio; Velha, Philippe; Pintus, Paolo;
Saponara, Sergio; Di Pasquale, Fabrizio; Oton, Claudio J.; Faralli, Stefano

被引用次数: 0

2024

On hardware acceleration of quantum-resistant FOTA systems in automotive

COMPUTERS & ELECTRICAL ENGINEERING

Perazzo, Pericle; Di Matteo, Stefano; Dini, Gianluca; Saponara, Sergio

被引用次数: 0

2024

Leaveraging Digital Twin & Artificial Intelligence in Consumption Forecasting System for Sustainable Luxury Yacht

IEEE ACCESS

Dini, Pierpaolo; Paolini, Davide; Minossi, Maurizio; Saponara, Sergio

被引用次数: 0

2024

HW-SW Interface Design and Implementation for Error Logging and Reporting for RAS Improvement

IEEE ACCESS

Canino, Nicasio; Di Matteo, Stefano; Rossi, Daniele; Saponara, Sergio

被引用次数: 0

2024

CRYPHTOR: A Memory-Unified NTT-Based Hardware Accelerator for Post-Quantum CRYSTALS Algorithms

IEEE ACCESS

Di Matteo, Stefano; Sarno, Ivan; Saponara, Sergio

被引用次数: 4

2024

TeleStroke: real-time stroke detection with federated learning and YOLOv8 on edge devices

JOURNAL OF REAL-TIME IMAGE PROCESSING

Elhanashi, Abdussalam; Dini, Pierpaolo; Saponara, Sergio; Zheng, Qinghe

被引用次数: 0

2024

SLOPE: Safety LOG PEripherals implementation and software drivers for a safe RISC-V microcontroller unit

MICROPROCESSORS AND MICROSYSTEMS

Cosimi, Francesco; Arena, Antonio; Saponara, Sergio; Gai, Paolo

被引用次数: 0

2024

Review on Modeling and SOC/SOH Estimation of Batteries for Automotive Applications

BATTERIES-BASEL

Dini, Pierpaolo; Colicelli, Antonio; Saponara, Sergio

被引用次数: 14

2024

Design and Experimental Verification of a 6.25-GHz PLL for Harsh Temperature Conditions in 65-nm CMOS Technology

IEEE TRANSACTIONS ON INSTRUMENTATION AND MEASUREMENT

Mestice, Marco; Ciarpi, Gabriele; Rossi, Daniele; Saponara, Sergio

被引用次数: 0

2024

Advancements in TinyML: Applications, Limitations, and Impact on IoT Devices

ELECTRONICS

Elhanashi, Abdussalam; Dini, Pierpaolo; Saponara, Sergio; Zheng, Qinghe

被引用次数: 0

2024

Overview of AI-Models and Tools in Embedded IIoT Applications

ELECTRONICS

Dini, Pierpaolo; Diana, Lorenzo; Elhanashi, Abdussalam; Saponara, Sergio

被引用次数: 2

2024

Edge HPC Architectures for AI-Based Video Surveillance Applications

ELECTRONICS

Rossi, Federico; Saponara, Sergio

被引用次数: 0

2024

An Integrated Charge Pump for Phase-Locked Loop Applications in Harsh Environments

ELECTRONICS

Mestice, Marco; Ciarpi, Gabriele; Rossi, Daniele; Saponara, Sergio

被引用次数: 1

2024

Real-time monitoring and ageing detection algorithm design with application on SiC-based automotive power drive system

IET POWER ELECTRONICS

Dini, Pierpaolo; Basso, Giovanni; Saponara, Sergio; Romano, Claudio

被引用次数: 4

2024

Modeling and Control Simulation of Power Converters in Automotive Applications

APPLIED SCIENCES-BASEL

Dini, Pierpaolo; Saponara, Sergio

被引用次数: 4

2024

RTL Flow for the Power Side-Channel Resilience Assessment of a Post-quantum SHA-3 Accelerator

2024 13TH INTERNATIONAL CONFERENCE ON MODERN CIRCUITS AND SYSTEMS TECHNOLOGIES, MOCAST 2024

Tenentes, Vasileios; Di Matteo, Stefano; Zonios, Christos; Rossi, Daniele; Saponara, Sergio

被引用次数: 0

2024

Real-Time Stroke Detection Using Deep Learning and Federated Learning

A real-time transformer discharge pattern recognition method based on CNN-LSTM driven by few-shot learning

ELECTRIC POWER SYSTEMS RESEARCH

Zheng, Qinghe; Wang, Ruoyu; Tian, Xinyu; Yu, Zhiguo; Wang, Hongjun; Elhanashi, Abdussalam; Saponara, Sergio

被引用次数: 7

2023

Integration of Deep Learning into the IoT: A Survey of Techniques and Challenges for Real-World Applications

ELECTRONICS

Elhanashi, Abdussalam; Dini, Pierpaolo; Saponara, Sergio; Zheng, Qinghe

被引用次数: 13

2023

An integrated and real-time social distancing, mask detection, and facial temperature video measurement system for pandemic monitoring

JOURNAL OF REAL-TIME IMAGE PROCESSING

Elhanashi, Abdussalam; Saponara, Sergio; Dini, Pierpaolo; Zheng, Qinghe; Morita, Daiki; Raytchev, Bisser

被引用次数: 7

2023

MobileRaT: A Lightweight Radio Transformer Method for Automatic Modulation Classification in Drone Communication Systems

DRONES

Zheng, Qinghe; Tian, Xinyu; Yu, Zhiguo; Ding, Yao; Elhanashi, Abdussalam; Saponara, Sergio; Kpalma, Kidiyo

被引用次数: 5

2023

A Real-Time Vehicle Speed Prediction Method Based on a Lightweight Informer Driven by Big Temporal Data

BIG DATA AND COGNITIVE COMPUTING

Tian, Xinyu; Zheng, Qinghe; Yu, Zhiguo; Yang, Mingqiang; Ding, Yao; Elhanashi, Abdussalam; Saponara, Sergio; Kpalma, Kidiyo

被引用次数: 3

2023

Overview on Intrusion Detection Systems Design Exploiting Machine Learning for Networking Cybersecurity

APPLIED SCIENCES-BASEL

Dini, Pierpaolo; Elhanashi, Abdussalam; Begni, Andrea; Saponara, Sergio; Zheng, Qinghe; Gasmi, Kaouther

被引用次数: 18

2023

DL-PR: Generalized automatic modulation classification method based on deep learning with priori regularization

ENGINEERING APPLICATIONS OF ARTIFICIAL INTELLIGENCE

Zheng, Qinghe; Tian, Xinyu; Yu, Zhiguo; Wang, Hongjun; Elhanashi, Abdussalam; Saponara, Sergio

被引用次数: 36

2023

Classification and Localization of Multi-Type Abnormalities on Chest X-Rays Images

IEEE ACCESS

Elhanashi, Abdussalam; Saponara, Sergio; Zheng, Qinghe

被引用次数: 6

2023

Experimental Characterization and Electro-Thermal Modeling of Double Side Cooled SiC MOSFETs for Accurate and Rapid Power Converter Simulations

IEEE ACCESS

Dini, Pierpaolo; Saponara, Sergio; Chakraborty, Sajib; Hosseinabadi, Farzad; Hegazy, Omar

被引用次数: 10

2023

A 10 Gb/s Line Driver in 65 nm CMOS Technology for Radiation-Pervaded and High-Temperature Applications

IEEE ACCESS

Ciarpi, Gabriele; Mestice, Marco; Rossi, Daniele; Palla, Fabrizio; Saponara, Sergio

被引用次数: 1

2023

VLSI Design and FPGA Implementation of an NTT Hardware Accelerator for Homomorphic SEAL-Embedded Library

IEEE ACCESS

Di Matteo, Stefano; Lo Gerfo, Matteo; Saponara, Sergio

被引用次数: 16

2023

Fast detection of bag-breakups in pulsating and steady airflow using video analysis and deep learning

JOURNAL OF REAL-TIME IMAGE PROCESSING

Morita, Daiki; Raytchev, Bisser; Elhanashi, Abdussalam; Kawaguchi, Mikimasa; Ogata, Yoichi; Higaki, Toru; Kaneda, Kazufumi; Nakashima, Akira; Saponara, Sergio

被引用次数: 1

2023

Design Methodology and Metrics for Robust and Highly Qualified Security Modules in Trusted Environments

ELECTRONICS

Crocetti, Luca; Nannipieri, Pietro; Di Matteo, Stefano; Saponara, Sergio

被引用次数: 0

2023

Overview on Battery Charging Systems for Electric Vehicles

ELECTRONICS

Dini, Pierpaolo; Saponara, Sergio; Colicelli, Antonio

被引用次数: 12

2023

Review of Methodologies and Metrics for Assessing the Quality of Random Number Generators

ELECTRONICS

Crocetti, Luca; Nannipieri, Pietro; Di Matteo, Stefano; Fanucci, Luca; Saponara, Sergio

被引用次数: 9

2023

Asymmetric 5.5 GHz Three-Stage Voltage-Controlled Ring-Oscillator in 65 nm CMOS Technology

ELECTRONICS

Ciarpi, Gabriele; Monda, Danilo; Mestice, Marco; Rossi, Daniele; Saponara, Sergio

被引用次数: 0

2023

Real-time electro-thermal modelling & predictive control design of resonant power converter in full electric vehicle applications

IET POWER ELECTRONICS

Dini, Pierpaolo; Ariaudo, Giorgio; Botto, Gianluca; La Greca, Francesco; Saponara, Sergio

被引用次数: 19

2023

Design and Experimental Assessment of Real-Time Anomaly Detection Techniques for Automotive Cybersecurity

SENSORS

Dini, Pierpaolo; Saponara, Sergio

被引用次数: 4

2023

From SW Timing Analysis and Safety Logging to HW Implementation: A Possible Solution with an Integrated and Low-Power Logger Approach

JOURNAL OF LOW POWER ELECTRONICS AND APPLICATIONS

Cosimi, Francesco; Arena, Antonio; Gai, Paolo; Saponara, Sergio

被引用次数: 1

2023

Implementation of a Hash-based Secure Core for Integrity and Authentication of Data in Space Applications on Space-grade FPGAs

2023 IEEE 2ND INDUSTRIAL ELECTRONICS SOCIETY ANNUAL ON-LINE CONFERENCE, ONCON

Crocetti, Luca; Falaschi, Francesco; Saponara, Sergio; Fanucci, Luca

被引用次数: 1

2023

Developing a real-time social distancing detection system based on YOLOv4-tiny and bird-eye view for COVID-19

JOURNAL OF REAL-TIME IMAGE PROCESSING

Saponara, Sergio; Elhanashi, Abdussalam; Zheng, Qinghe

被引用次数: 14

2022

Fine-Grained Modulation Classification Using Multi-Scale Radio Transformer With Dual-Channel Representation

IEEE COMMUNICATIONS LETTERS

Zheng, Qinghe; Zhao, Penghui; Wang, Hongjun; Elhanashi, Abdussalam; Saponara, Sergio

被引用次数: 66

2022

Design and Test of an Integrated Random Number Generator with All-Digital Entropy Source

ENTROPY

Crocetti, Luca; Di Matteo, Stefano; Nannipieri, Pietro; Fanucci, Luca; Saponara, Sergio

被引用次数: 11

2022

Polarization Control in Integrated Graphene-Silicon Quantum Photonics Waveguides

MATERIALS

Cammarata, Simone; Fontana, Andrea; Kaplan, Ali Emre; Cornia, Samuele; Thu Ha Dao; Lacava, Cosimo; Demontis, Valeria; Iadanza, Simone; Vitali, Valerio; De Matteis, Fabio; Pedreschi, Elena; Magazzu, Guido; Toncelli, Alessandra; Spinella, Franco; Saponara, Sergio; Gunnella, Roberto; Rossella, Francesco; Salamon, Andrea; Bellani, Vittorio

被引用次数: 3

2022

A Lightweight Posit Processing Unit for RISC-V Processors in Deep Neural Network Applications

IEEE TRANSACTIONS ON EMERGING TOPICS IN COMPUTING

Cococcioni, Marco; Rossi, Federico; Ruffaldi, Emanuele; Saponara, Sergio

被引用次数: 15

2022

Design and Testing Novel One-Class Classifier Based on Polynomial Interpolation With Application to Networking Security

IEEE ACCESS

Dini, Pierpaolo; Begni, Andrea; Ciavarella, Stefano; De Paoli, Emiliano; Fiorelli, Giuseppe; Silvestro, Carmelo; Saponara, Sergio

被引用次数: 28

2022

Processor-in-the-Loop Validation of a Gradient Descent-Based Model Predictive Control for Assisted Driving and Obstacles Avoidance Applications

IEEE ACCESS

Dini, Pierpaolo; Saponara, Sergio

被引用次数: 32

2022

Towards EXtreme scale technologies and accelerators for euROhpc hw/Sw supercomputing applications for exascale: The TEXTAROSSA approach

MICROPROCESSORS AND MICROSYSTEMS

Agosta, Giovanni; Aldinucci, Marco; Alvarez, Carlos; Ammendola, Roberto; Arfat, Yasir; Beaumont, Olivier; Bernaschi, Massimo; Biagioni, Andrea; Boccali, Tommaso; Bramas, Berenger; Brandolese, Carlo; Cantalupo, Barbara; Carrozzo, Mauro; Cattaneo, Daniele; Celestini, Alessandro; Celino, Massimo; Colonnelli, Iacopo; Cretaro, Paolo; D'Ambra, Pasqua; Danelutto, Marco; Esposito, Roberto; Eyraud-Dubois, Lionel; Filgueras, Antonio; Fornaciari, William; Freza, Ottorino; Galimberti, Andrea; Giacomini, Francesco; Goglin, Brice; Gregori, Daniele; Guermouche, Abdou; Iannone, Francesco; Kulczewski, Michal; Lo Cicero, Francesca; Lonardo, Alessandro; Martinelli, Alberto R.; Martinelli, Michele; Martorell, Xavier; Massari, Giuseppe; Montangero, Simone; Mittone, Gianluca; Namyst, Raymond; Oleksiak, Ariel; Palazzari, Paolo; Paolucci, Pier Stanislao; Reghenzani, Federico; Rossi, Cristian; Saponara, Sergio; Simula, Francesco; Terraneo, Federico; Thibault, Samuel; Torquati, Massimo; Turisini, Matteo; Vicini, Piero; Vidal, Miquel; Zoni, Davide; Zummo, Giuseppe

被引用次数: 3

2022

Design and Characterization of 10 Gb/s and 1 Grad TID-Tolerant Optical Modulator Driver

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Ciarpi, Gabriele; Cammarata, Simone; Monda, Danilo; Faralli, Stefano; Velha, Philippe; Magazzu, Guido; Palla, Fabrizio; Saponara, Sergio

被引用次数: 5

2022

VLSI Design of Advanced-Features AES Cryptoprocessor in the Framework of the European Processor Initiative

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Nannipieri, Pietro; Matteo, Stefano Di; Baldanzi, Luca; Crocetti, Luca; Zulberti, Luca; Saponara, Sergio; Fanucci, Luca

被引用次数: 11

2022

A Script-Based Cycle-True Verification Framework to Speed-Up Hardware and Software Co-Design: Performance Evaluation on ECC Accelerator Use-Case

ELECTRONICS

Zulberti, Luca; Di Matteo, Stefano; Nannipieri, Pietro; Saponara, Sergio; Fanucci, Luca

被引用次数: 9

2022

Electro-Thermal Model-Based Design of Bidirectional On-Board Chargers in Hybrid and Full Electric Vehicles

ELECTRONICS

Dini, Pierpaolo; Saponara, Sergio

被引用次数: 28

2022

Review on Model Based Design of Advanced Control Algorithms for Cogging Torque Reduction in Power Drive Systems

ENERGIES

Dini, Pierpaolo; Saponara, Sergio

被引用次数: 12

2022

On the Hardware-Software Integration in Cryptographic Accelerators for Industrial IoT

APPLIED SCIENCES-BASEL

Leonardi, Luigi; Lettieri, Giuseppe; Perazzo, Pericle; Saponara, Sergio

被引用次数: 2

2022

Recreating Fingerprint Images by Convolutional Neural Network Autoencoder Architecture

IEEE ACCESS

Saponara, Sergio; Elhanashi, Abdussalam; Zheng, Qinghe

被引用次数: 9

2021

Guest Editorial: Special issue on parallel, distributed, and network-based processing in next-generation embedded systems

JOURNAL OF SYSTEMS ARCHITECTURE

Mubeen, Saad; Lo Bello, Lucia; Daneshtalab, Masoud; Saponara, Sergio

被引用次数: 1

2021

Time-Sensitive Networking in automotive embedded systems: State of the art and research opportunities

JOURNAL OF SYSTEMS ARCHITECTURE

Ashjaei, Mohammad; Lo Bello, Lucia; Daneshtalab, Masoud; Patti, Gaetano; Saponara, Sergio; Mubeen, Saad

被引用次数: 75

2021

Vectorizing posit operations on RISC-V for faster deep neural networks: experiments and comparison with ARM SVE

NEURAL COMPUTING & APPLICATIONS

Cococcioni, Marco; Rossi, Federico; Ruffaldi, Emanuele; Saponara, Sergio

被引用次数: 9

2021

A RISC-V Post Quantum Cryptography Instruction Set Extension for Number Theoretic Transform to Speed-Up CRYSTALS Algorithms

IEEE ACCESS

Nannipieri, Pietro; Di Matteo, Stefano; Zulberti, Luca; Albicocchi, Francesco; Saponara, Sergio; Fanucci, Luca

被引用次数: 36

2021

A real-time video smoke detection algorithm based on Kalman filter and CNN

JOURNAL OF REAL-TIME IMAGE PROCESSING

Gagliardi, Alessio; de Gioia, Francesco; Saponara, Sergio

被引用次数: 20

2021

Implementing a real-time, AI-based, people detection and social distancing measuring system for Covid-19

JOURNAL OF REAL-TIME IMAGE PROCESSING

Saponara, Sergio; Elhanashi, Abdussalam; Gagliardi, Alessio

被引用次数: 48

2021

Real-time video fire/smoke detection based on CNN in antifire surveillance systems

JOURNAL OF REAL-TIME IMAGE PROCESSING

Saponara, Sergio; Elhanashi, Abdussalam; Gagliardi, Alessio

被引用次数: 107

2021

SHA2 and SHA-3 accelerator design in a 7 nm technology within the European Processor Initiative

MICROPROCESSORS AND MICROSYSTEMS

Nannipieri, Pietro; Bertolucci, Matteo; Baldanzi, Luca; Crocetti, Luca; Di Matteo, Stefano; Falaschi, Francesco; Fanucci, Luca; Saponara, Sergio

被引用次数: 15

2021

Novel Arithmetics in Deep Neural Networks Signal Processing for Autonomous Driving: Challenges and Opportunities

IEEE SIGNAL PROCESSING MAGAZINE

Cococcioni, Marco; Rossi, Federico; Ruffaldi, Emanuele; Saponara, Sergio; de Dinechin, Benoit Dupont

被引用次数: 21

2021

Design and Verification of a 6.25 GHz LC-Tank VCO Integrated in 65 nm CMOS Technology Operating up to 1 Grad TID

IEEE TRANSACTIONS ON NUCLEAR SCIENCE

Monda, Danilo; Ciarpi, Gabriele; Saponara, Sergio

被引用次数: 12

2021

Model-Based Design of an Improved Electric Drive Controller for High-Precision Applications Based on Feedback Linearization Technique

ELECTRONICS

Dini, Pierpaolo; Saponara, Sergio

被引用次数: 25

2021

Diagnosis of Faults Induced by Radiation and Circuit-Level Design Mitigation Techniques: Experience from VCO and High-Speed Driver CMOS ICs Case Studies

ELECTRONICS

Monda, Danilo; Ciarpi, Gabriele; Saponara, Sergio

被引用次数: 2

2021

Secure Elliptic Curve Crypto-Processor for Real-Time IoT Applications

ENERGIES

Di Matteo, Stefano; Baldanzi, Luca; Crocetti, Luca; Nannipieri, Pietro; Fanucci, Luca; Saponara, Sergio

被引用次数: 26

2021

Power Converters, Electric Drives and Energy Storage Systems for Electrified Transportation and Smart Grid ApplicationsF

ENERGIES

Mihet-Popa, Lucian; Saponara, Sergio

被引用次数: 4

2021

Optimized Energy and Air Quality Management of Shared Smart Buildings in the COVID-19 Scenario

ENERGIES

Anastasi, Giuseppe; Bartoli, Carlo; Conti, Paolo; Crisostomi, Emanuele; Franco, Alessandro; Saponara, Sergio; Testi, Daniele; Thomopoulos, Dimitri; Vallati, Carlo

被引用次数: 16

2021

An Instrument for the Characterization and Calibration of Optical Sensors

SENSORS

Gastasini, Enrico; Capecci, Niccolo; Lupi, Francesco; Gagliardi, Alessio; Saponara, Sergio; Lanzetta, Michele

被引用次数: 5

2021

Recent Trends on IoT Systems for Traffic Monitoring and for Autonomous and Connected Vehicles

SENSORS

Saponara, Sergio; Giordano, Stefano; Mariani, Riccardo

被引用次数: 0

2021

Performance Evaluation of Attribute-Based Encryption in Automotive Embedded Platform for Secure Software Over-The-Air Update

SENSORS

La Manna, Michele; Treccozzi, Luigi; Perazzo, Pericle; Saponara, Sergio; Dini, Gianluca

被引用次数: 21

2021

True Random Number Generator Based on Fibonacci-Galois Ring Oscillators for FPGA

APPLIED SCIENCES-BASEL

Nannipieri, Pietro; Di Matteo, Stefano; Baldanzi, Luca; Crocetti, Luca; Belli, Jacopo; Fanucci, Luca; Saponara, Sergio

被引用次数: 24

2021

TEXTAROSSA: Towards EXtreme scale Technologies and Accelerators for euROhpc hw/Sw Supercomputing Applications for exascale

2021 24TH EUROMICRO CONFERENCE ON DIGITAL SYSTEM DESIGN (DSD 2021)

Agosta, Giovanni; Cattaneo, Daniele; Fornaciari, William; Galimberti, Andrea; Massari, Giuseppe; Reghenzani, Federico; Terraneo, Federico; Zoni, Davide; Brandolese, Carlo; Celino, Massimo; Iannone, Francesco; Palazzari, Paolo; Zummo, Giuseppe; Bernaschi, Massimo; D'Ambra, Pasqua; Saponara, Sergio; Danelutto, Marco; Torquati, Massimo; Aldinucci, Marco; Arfat, Yasir; Cantalupo, Barbara; Colonnelli, Iacopo; Esposito, Roberto; Martinelli, Alberto R.; Mittone, Gianluca; Beaumont, Olivier; Bramas, Berenger; Eyraud-Dubois, Lionel; Goglin, Brice; Guermouche, Abdou; Namyst, Raymond; Thibault, Samuel; Filgueras, Antonio; Vidal, Miquel; Alvarez, Carlos; Martorell, Xavier; Oleksiak, Ariel; Kulczewski, Michal; Lonardo, Alessandro; Vicini, Piero; Lo Cicero, Francesca; Simula, Francesco; Biagioni, Andrea; Cretaro, Paolo; Frezza, Ottorino; Paolucci, Pier Stanislaw; Turisini, Matteo; Giacomini, Francesco; Boccali, Tommaso; Montangero, Simone; Ammendola, Roberto

被引用次数: 5

2021

ELECTRIC DRIVES AND POWER CHARGERS Recent Solutions to Improve Performance and Energy Efficiency for Hybrid and Fully Electric Vehicles

IEEE VEHICULAR TECHNOLOGY MAGAZINE

Saponara, Sergio; Lee, Christopher H. T.; Wang, Nelson Xuntuo; Kirtley, James L., Jr.

被引用次数: 35

2020

Fast deep neural networks for image processing using posits and ARM scalable vector extension

JOURNAL OF REAL-TIME IMAGE PROCESSING

Cococcioni, Marco; Rossi, Federico; Ruffaldi, Emanuele; Saponara, Sergio

被引用次数: 10

2020

Design, Implementation, and Experimental Verification of 5 Gbps, 800 Mrad TID and SEU-Tolerant Optical Modulators Drivers

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Ciarpi, Gabriele; Magazzu, Guido; Palla, Fabrizio; Saponara, Sergio

被引用次数: 9

2020

Formal Verification and Co-Simulation in the Design of a Synchronous Motor Control Algorithm

ENERGIES

Bernardeschi, Cinzia; Dini, Pierpaolo; Domenici, Andrea; Palmieri, Maurizio; Saponara, Sergio

被引用次数: 25

2020

Design and Performance Evaluation of Multi-Gb/s Silicon Photonics Transmitters for High Energy Physics

ENERGIES

Cammarata, Simone; Ciarpi, Gabriele; Faralli, Stefano; Velha, Philippe; Magazzu, Guido; Palla, Fabrizio; Saponara, Sergio

被引用次数: 4

2020

Design of Adaptive Controller Exploiting Learning Concepts Applied to a BLDC-Based Drive System

ENERGIES

Dini, Pierpaolo; Saponara, Sergio

被引用次数: 30

2020

AdViSED: Advanced Video Smoke Detection for Real-Time Measurements in Antifire Indoor and Outdoor Systems

ENERGIES

Gagliardi, Alessio; Saponara, Sergio

被引用次数: 16

2020

Integrated Passive Devices and Switching Circuit Design for a 3D DC/DC Converter up to 60 V

JOURNAL OF CIRCUITS SYSTEMS AND COMPUTERS

Saponara, Sergio; Ciarpi, Gabriele; Erlbacher, Tobias; Rattmann, Gudrun

被引用次数: 1

2020

Recent Trends on Applications of Electronics Pervading the Industry, Environment and Society

SENSORS

Saponara, Sergio; De Gloria, Alessandro; Bellotti, Francesco

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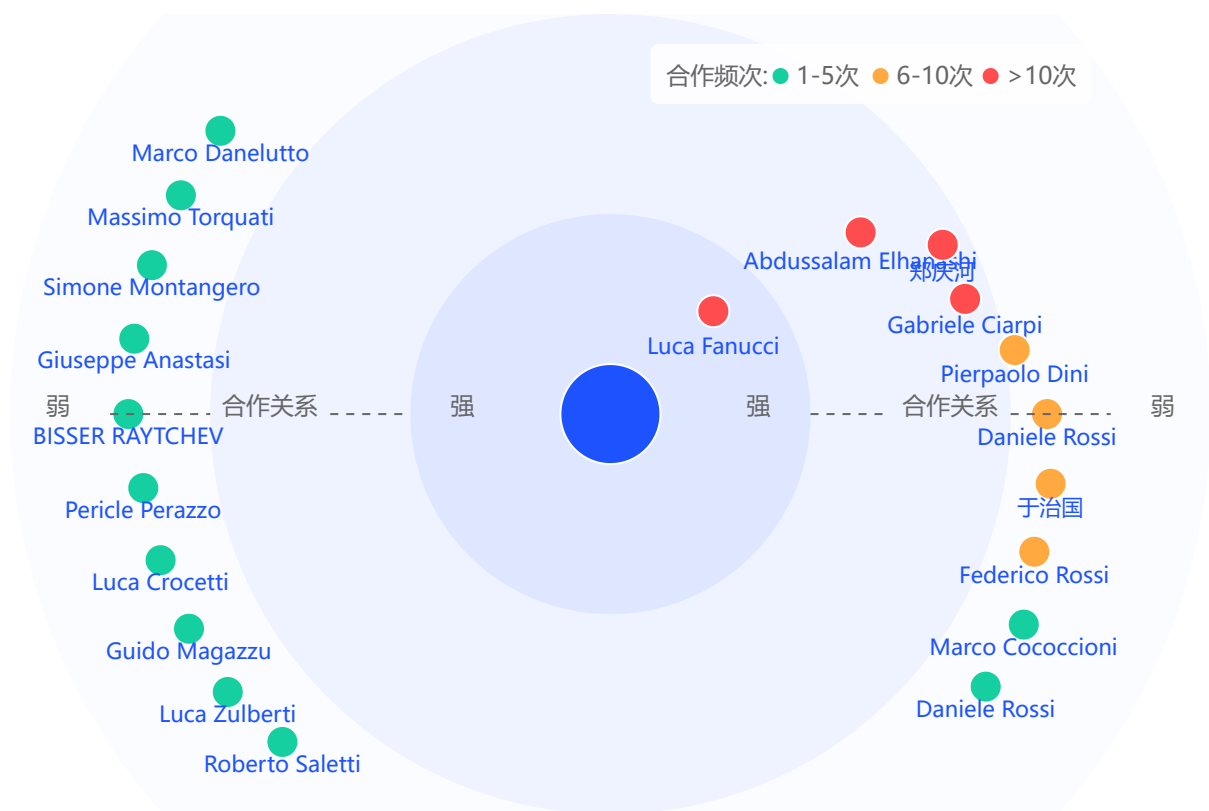
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合作次数: 39

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